

# FASERCAL V10

Install, Build, Run — a consolidated reference

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# Outline

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- **ConvertGENIE / ConvertFASERMC**: convert a GENIE or official FASERMC neutrino-interaction sample into the common `TP0Event` format
- **FASERG4** (`faserps`): runs that through Geant4, writes per-event `TcalEvent` truth files
- **Batch** (`batchreco.exe`): reconstructs truth files into `TP0RecoEvent` files
- **EvDisplay** / analysis / **TauSearch**: consumes the reconstructed output
- `run_regression_tests.py`: wraps the whole chain end to end for an automated simulate → reconstruct → compare check

## Known issue: FASERG4 scintillator proton fraction ( $V9 \rightarrow V10$ )

- FASERG4's scintillator material definition (3DCal tracker layers) had the wrong free-proton (H) vs. carbon-12 (C) content — fixed between production versions **V9** and **V10**
- **Root cause:** `G4Material::AddElement()` is overloaded — one variant takes an atom count (`G4int`), another a mass fraction (`G4double`). Declaring the stoichiometry counters as `G4double` silently selected the mass-fraction overload, so atom counts were read as mass fractions
- **Symptom:** free-proton and carbon-12 targets interacted at almost the same rate in V9 ( $\approx 1:1$ ) instead of the physically expected  $\approx 1:12$  ( $\propto$  mass number  $A$ ); V10 correctly gives  $\approx 12:1$
- **Effect:** neutrino-induced CC/NC rates rise  $\sim 15\text{--}18\%$  (a more isoscalar target supplies more neutrons for the neutrino-preferred  $d$ -quark coupling), antineutrino rates stay roughly flat; overall 3DCal fiducial total  $+4.4\%$
- also shifts scintillator density ( $1.03 \rightarrow 1.06 \text{ g/cm}^3$ ), hadronic interaction length, and  $dE/dx$  ( $Z/A$  change); Birks' constant has *not* yet been re-checked against the new composition
- full analysis: `FASERflux/faserg4_proton_fraction_bug.tex` (Fabio Cufino, André Rubbia)

# Known issue: muon-spectrometer field consistency

- **Motivation:** the muon spectrometer's magnetic field was implemented *twice* — once in Geant4 simulation (`MuonMagneticField`) and once in GenFit reconstruction (`GenMagneticField`) — two independent hand-written implementations kept in sync only by hand, with no test proving they agreed and no check that either matched the real detector geometry
- **Fix:** extracted one shared, framework-independent function, `FASER::ComputeMuonSpectrometerField()`; both G4 and GenFit now call it as thin unit-converting adapters, so the two field *formulas* can no longer silently diverge — `MuonSpectrometerFieldTest (gtest)` checks G4 and GenFit agree
- **Real bug caught by the test:** outside the field's modelled envelope, GenFit's plain “Magnet” branch fell through to a stray non-zero fallback (0.001 kG in *Y*) where G4 returned an exact zero — a genuine, if tiny, pre-existing sim/reco divergence
- **Second problem:** the slit position deciding where the field flips sign / cuts to zero was a bare hardcoded literal duplicated in **four** places, agreeing only by convention, not construction — a future geometry change could silently desynchronize the field model from the geometry
- **Fix:** `MagnetGeometryProbe` samples the *actual* `TGeoShape` loaded from the real GDML to find where the iron truly ends and the slit truly begins, so reconstruction derives the slit position live instead of trusting a literal; `MagnetGeometryProbeTest` unit-tests the probe itself, including a case that deliberately mismatches the geometry to prove a future drift would actually be caught

# 1. Install from scratch

```
git clone https://github.com/rubbiaa/FASER.git
cd FASER
source setup.sh
```

- `setup.sh` auto-detects the site (André's Mac, the Ubuntu ryzen01 box, or lxplus) and sources the matching `thisroot.sh` / `geant4.sh`
- hands off to `common_setup.sh`: CLHEP/Rave/GenFit paths, `LD_LIBRARY_PATH`, `CMAKE_PREFIX_PATH`, `PATH`, and a sanity check for missing/stale variables
- defines `fb` ("faser build"): reruns `cmake -build $HOMEFASER/build -j` from anywhere
- unrecognized machine? it prints exactly which `elif` branch to add
- full from-scratch ROOT/Geant4 install: `docs/INSTALL.md`, `docs/GEANT4_INSTALL.md`

## 2. Build

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=RelWithDebInfo
cmake --build build -j
```

- first build also compiles CLHEP, Rave, GenFit, googletest, and Pythia8 (CMake superbuild) — slow once, incremental afterward
- after source `setup.sh`, `fb` reruns the build step from anywhere
- executables land in `build/bin/`: `faserps`, `batchreco.exe`, `batchreco_detresp.exe`, `dumphits.exe`, `evDisplay.exe`, `ConvertGENIE.exe`, `Convert.exe`, `CombineFluxes.exe`, the `gtest` binaries, ...

# CMake build options

| Option                      | Default | Meaning                                                          |
|-----------------------------|---------|------------------------------------------------------------------|
| FASER_BUILD_EXTERNALS       | ON      | Build CLHEP/Rave/GenFit/ googletest/Pythia8 from source          |
| FASER_BUILD_GEANT4_PACKAGES | ON      | Build FASERG4 and FASERCALProtoG4 (needs Geant4)                 |
| FASER_BUILD_DISPLAY         | ON      | Build the GenFit-based event display / muon spectrometer package |
| FASER_BUILD_TAUSEARCH       | ON      | Build TauSearch                                                  |
| FASER_BUILD_TESTS           | ON      | Build FASERCAL's own gtest suite, registered with CTest          |

Full list and reuse-an-existing-dependency examples: docs/INSTALL.md

### 3. Repository layout

FASER/ is the git repo (FASERCAL is this project's own name — see README.md); top level also holds CMakeLists.txt and setup.sh/common\_setup.sh.

- CoreUtils/ — core library
- ConvertGENIE/ — GENIE → TPOEvent
- ConvertFASERMC/ — FASERMC → TPOEvent
- ConvertNPZ/ — NPZ conversion
- FASERG4/ — faserps (Geant4 sim)
- FASERCALProtoG4/ — proto calo G4
- Batch/ — batchreco.exe
- Display/ — event display / muon spec
- EvDisplay/ — evDisplay.exe
- Analysis/ — post-reco analysis
- TauSearch/ — tau-neutrino search
- FileMask/ — interaction-type masks
- Tests/ — FASERCAL's gtest suite
- cmake/ — CMake helper modules
- docs/ — this deck + docs
- data/ — default \$FASERDATA
- build/ — CMake build dir
- run\_\*.py, fetch\_data.py — wrappers

All gitignored except source; data/, build/ are generated. Analysis/, Display/, EvDisplay/, TauSearch/, FASERCALProtoG4/ each gate on their own FASER\_BUILD\_\* CMake option.

## 4. \$FASERDATA

- every FASERG4/Batch executable resolves its data location through \$FASERDATA (FASER::GetDataDir(), in CoreUtils/FaserDataDir.hh) — not a hardcoded relative path or an inter-directory symlink
- GetDataDir() creates a requested subdirectory on demand if it doesn't already exist
- common\_setup.sh defaults FASERDATA to \$HOMEFASER/data (gitignored) unless a per-site branch in setup.sh already exported it elsewhere (scratch, EOS, a dedicated checkout, ...)

## \$FASERDATA subdirectories

| Subdirectory          | Contents                                                           | Written by / Read by                                                           |
|-----------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------|
| faserG4/              | Per-event TcalEvent truth files                                    | faserps / batchreco.exe, dumphits.exe, evDisplay.exe                           |
| batch/                | Reconstructed TPORecoEvent files                                   | batchreco.exe / analysis code, summarize_output.py                             |
| GENIE/                | GENIE input samples (too large for git)                            | fetch_data.py / ConvertGENIE.exe (via run_convertgenie.py)                     |
| GDML/                 | Detector geometry, exported every run                              | faserps / run_batchreco.py's default --geometry-file                           |
| CVGENIE/<br>Run<run>/ | Per-run ConvertGENIE.exe output: PO ROOT files, logs, overlay PNGs | ConvertGENIE.exe (via run_convertgenie.py); then run_faserps.py's --input-file |

## 4. `run_convertgenie.py` — runs ConvertGENIE.exe per detector

- drives ConvertGENIE.exe (`-r <run> -g <gdmlfile> [-l <luminosity_fb>] <genieroot...> [detector]`), auto-discovering its GENIE input files (`$FASERDATA/GENIE/`) and GDML geometry (`$FASERDATA/GDML/`) so normally only `--run` needs to be given by hand
- by default runs **three times** — 3DCAL, ECAL, AHCAL — so all three detectors' PO-format output exists side by side instead of the last run clobbering the others
- output goes to `$FASERDATA/CVGENIE/Run<run>/`: one PO ROOT file per detector, a per-detector log plus a summary log with event counts by interaction category (nueCC, numuCC, nutauCC, NC, ES)

# run\_convertgenie.py flags

| Flag                      | Default                          | Meaning                                              |
|---------------------------|----------------------------------|------------------------------------------------------|
| --run                     | <i>required</i>                  | Run number, passed as ConvertGENIE.exe -r            |
| --detectors               | 3DCAL ECAL AHCAL                 | Which detector(s) to run; also accepts MuonSpec, ALL |
| --input-files             | auto-discovered                  | GENIE *.gfaser.root file(s)/patterns                 |
| --genie-dir               | \$FASERDATA/GENIE                | Where to auto-discover --input-files from            |
| --geometry-file           | auto-discovered                  | GDMML geometry passed as ConvertGENIE.exe -g         |
| --charmonly / --taucconly | off                              | Passed through as -charmonly / -tauCConly            |
| --build-dir               | build/                           | CMake build dir containing bin/ConvertGENIE.exe      |
| --output-dir              | \$FASERDATA/CVGENIE/<br>Run<run> | Where PO files, logs, plots are written              |
| --dry-run                 | off                              | Print the command(s), don't execute                  |
| --force / -f              | off                              | Skip the confirmation prompt, delete+regenerate      |
| --no-overlay              | off                              | Skip the cross-detector overlay-PNG step             |

## run\_convertgenie.py behavior

- **luminosity**: auto-detected by scanning the GENIE production's own `run*.sh` script(s) for the `gevgen_faser -l <value>` it was generated with; recorded in the summary log (sets the scale the event counts should be read against) and passed to `ConvertGENIE.exe -l`, which also puts it in the reference-plot titles
- **re-running**: if `--output-dir` already has files in it, asks for confirmation before wiping it and regenerating from scratch (`--force/-f` skips the prompt); either way the directory is removed completely first, so no stale files linger
- **overlay step**: after all detectors finish, runs `ConvertGENIE/make_overlay_plots.C` (skippable with `--no-overlay`; skipped with a warning, not a failure, if `root` isn't on `PATH`) to produce one overlay PNG per quantity with every run detector superimposed

## `run_faserps.py` — runs the Geant4 simulation

- builds the V10-geometry macro in memory and pipes it straight to `faserps`' stdin — no `.mac` file to keep in sync
- full prose docs (custom geometry, interactive UI, MuonDIS physics): `run_faserps.md`
- default input sample is auto-discovered from `$FASERDATA/CVGENIE/Run<run>/` (written by `run_convertgenie.py`), prompting for run/detector if more than one is available

# run\_faserps.py flags (1/2)

| Flag          | Default         | Meaning                                                            |
|---------------|-----------------|--------------------------------------------------------------------|
| --build-dir   | build/          | CMake build dir containing bin/faserps                             |
| --vis         | off             | Launch the interactive Geant4 UI (no macro used)                   |
| --print-macro | off             | Print the resolved macro text, exit                                |
| --dry-run     | off             | Print the command, don't execute                                   |
| --input-file  | auto-discovered | ROOT input file; a relative custom value resolves against FASERG4/ |
| --run         | auto-discovered | Which CVGENIE/Run<run>/ to read from (skips the run prompt)        |
| --detector    | auto-discovered | 3DCAL/AHCAL/ ECAL (skips the detector prompt)                      |
| --start-event | 0               | First event index to read                                          |
| --n-events    | 100             | Number of events (/run/beam0n)                                     |
| --muons       | off             | Generate fixed-momentum muons instead of reading --input-file      |

# run\_faserps.py flags (2/2)

| Flag                         | Default             | Meaning                                                              |
|------------------------------|---------------------|----------------------------------------------------------------------|
| --muon-momentum-gev          | 100.0               | Muon momentum [GeV]                                                  |
| --muondis                    | off                 | Enable on-the-fly Pythia8 DIS per muon interaction (implies --muons) |
| --muondis-cross-section-bias | 150.0               | Cross-section bias (1 = unbiased)                                    |
| --muondis-q2min              | 1.0                 | Minimum $Q^2$ [GeV <sup>2</sup> ]                                    |
| --muondis-interaction-log    | disabled            | CSV path logging each interaction                                    |
| --muondis-pdf-set            | built-in proton PDF | Path under FASERG4/input/                                            |
| --muondis-xbjmin             | 0.0                 | Minimum Bjorken-x cut                                                |
| --muondis-debug              | off                 | Verbose MuonDIS debug output                                         |
| --seed                       | 123456789           | Geant4 random seed (matches faserps.cc's hardcoded default)          |
| --tilt-deg                   | -4.5                | Detector tilt about Y [deg]                                          |
| --shift-x-cm / --shift-y-cm  | 45.0 / 24.0         | LOS shift [cm]                                                       |

## run\_batchreco.py — runs the reconstruction

- wraps Batch/batchreco.exe with named, validated flags instead of positional arguments
- full prose docs: run\_batchreco.md

| Flag                      | Default                                | Meaning                                         |
|---------------------------|----------------------------------------|-------------------------------------------------|
| --run                     | <i>required</i>                        | Run number — deliberately no default            |
| --min-event / --max-event | 0 / 99999999                           | Event index range [min, max)                    |
| --mask                    | none (all events)                      | nueCC, numuCC, nutauCC, nuNC, or nuES           |
| --multi-thread            | off                                    | Pass -mt to batchreco.exe                       |
| --geometry-file           | \$FASERDATA/GDML/<br>FASERCAL_V10.gdml | GDML geometry to load                           |
| --build-dir               | build/                                 | CMake build dir containing bin/batchreco.exe    |
| --split                   | 1                                      | Split the range into N parallel background jobs |
| --dry-run                 | off                                    | Print the command(s), don't execute             |

## run\_regression\_tests.py — simulate → reconstruct → compare

- runs the real pipeline end to end for six golden cases (nueCC/numuCC/nutauCC/nuNC, plus --muons and --muondis)
- compares every field (mean\_\*/rms\_\*, truth *and* reco) against a committed golden JSON baseline — --rel-tol  $10^{-9}$  by default: near machine precision, *not* a statistical tolerance (this suite expects exact reproducibility, see docs/REGRESSION\_TESTS.md)
- batchreco.exe's -mt is **on by default** here (verified against the single-threaded goldens first); raw faserps/batchreco.exe output no longer floods the terminal — it's redirected to work/<case>/logs/\*.log

```
python3 run_regression_tests.py --record # first time / deliberate change  
python3 run_regression_tests.py # every other time: compare
```

## run\_regression\_tests.py flags (1/2)

| Flag                     | Default                                       | Meaning                                                                                                                                                       |
|--------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--case</code>      | all cases                                     | Restrict to a golden case name (repeatable); see <code>--list</code>                                                                                          |
| <code>--record</code>    | off                                           | Write the golden file(s) instead of comparing — refuses to overwrite an existing one, see <code>--force</code>                                                |
| <code>--force</code>     | off                                           | Required with <code>--record</code> to overwrite a golden file that already exists (local backstop — <code>.github/CODEOWNERS</code> is the real review gate) |
| <code>--rel-tol</code>   | $10^{-9}$                                     | Relative tolerance for floating-point comparisons — near machine precision, not a statistical tolerance                                                       |
| <code>--build-dir</code> | build/                                        | CMake build directory                                                                                                                                         |
| <code>--python</code>    | <code>\$FASER_PYTHON</code> env, else current | Interpreter for the subprocess calls — needs a PyROOT-enabled one matching your ROOT build                                                                    |
| <code>--list</code>      | —                                             | List available case names, exit                                                                                                                               |

## run\_regression\_tests.py flags (2/2)

| Flag                                                         | Default  | Meaning                                                                                                                                                                                            |
|--------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>--skip-faserps</code>                                  | off      | Reuse the truth sample already in <code>work/&lt;case&gt;/data/faserG4</code> instead of re-simulating                                                                                             |
| <code>--skip-reco</code>                                     | off      | Reuse the reco file already in <code>work/&lt;case&gt;/data/batch</code> instead of re-running <code>batchreco.exe</code> ; combine with <code>--skip-faserps</code> to jump straight to summarize |
| <code>--multi-thread</code> / <code>--no-multi-thread</code> | on / off | Pass <code>batchreco.exe</code> 's <code>-mt</code> ( <b>on by default</b> here, unlike <code>run_batchreco.py</code> standalone)                                                                  |

Tests/regression/results/<case>.json (the real numbers) and, for a plain comparison, results/comparison.csv — one row per field per case: golden, current, abs\_diff, rel\_diff, status. See docs/REGRESSION\_TESTS.md's "Example: what a typical run looks like".

# Golden JSON schema

One file per case (Tests/regression/golden/<case>.json): meta records how it was produced; truth/reco are aggregate numbers only — every mean\_\* has an rms\_\* partner (ROOT's TH1::GetRMS() convention) — not full per-event dumps, so a diff stays readable in a PR. Real content of golden/neutrino\_numuCC.json (rounded/trimmed for space):

```
{
  "meta": {"case": "neutrino_numuCC", "reaction": "numuCC",
    "faserps_args": [], "run_number": 10000,
    "n_events_simulated": 100},
  "truth": {"n_events": 25,
    "mean_Evis": 595.73, "rms_Evis": 401.11,
    "mean_Q2": 77.26, "rms_Q2": 58.73,
    "mean_xBj": 0.202, "rms_xBj": 0.140, ...},
  "reco": {"n_events_reconstructed": 25,
    "mean_n_PORecs": 14.76, "rms_n_PORecs": 4.91,
    "mean_total_Ecompensated": 72.63,
    "rms_total_Ecompensated": 66.27, ...}
}
```

n\_events/n\_events\_reconstructed are how many of the 100 simulated events were actually this reaction (unbiased sample, see above) — not 100. muons/muondis omit the DIS fields entirely where meaningless, rather than padding with zeros.

## Example: what a typical run looks like

```
[run_regression_tests] neutrino: faserps: ... --input-file ... --n-events 100 ...
[run_regression_tests] neutrino: faserps: output -> work/neutrino/logs/faserps.log
[run_regression_tests] neutrino: batchreco: ... --multi-thread
[run_regression_tests] neutrino: batchreco: output -> work/neutrino/logs/batchreco.log
[run_regression_tests] neutrino_nueCC: PASS
[run_regression_tests] neutrino_numuCC: PASS
[run_regression_tests] neutrino_nutauCC: PASS
[run_regression_tests] neutrino_nuNC: PASS
[run_regression_tests] muons: PASS
[run_regression_tests] muondis: PASS
[run_regression_tests] full per-case results ... written under Tests/regression/results/
[run_regression_tests] field-by-field comparison ... (148 fields, 0 FAIL) written to
    Tests/regression/results/comparison.csv
```

Exit code 0 (every case PASS). A real FAIL example and real comparison.csv rows (golden, current, abs\_diff, rel\_diff, status — per field, not just the FAILs): docs/REGRESSION\_TESTS.md, §“Example: what a typical run looks like”.

## fetch\_data.py & summarize\_output.py

`fetch_data.py` — public-CERNBox-link downloads into `$FASERDATA`, sha256-verified. Manual step, not auto-triggered (the old auto-fetch-on-missing-file path was removed when GENIE/ became CVGENIE/); covers only the *raw* GENIE output, not the converted PO sample — see `docs/REGRESSION_TESTS.md`'s open items.

| Flag                   | Default | Meaning                                 |
|------------------------|---------|-----------------------------------------|
| <code>--force</code>   | off     | Re-download even if already present     |
| <code>--dry-run</code> | off     | Show what would be fetched, don't fetch |
| <code>--list</code>    | —       | List manifest entries, exit             |

`Tests/regression/summarize_output.py` — PyROOT summarizer used internally by `run_regression_tests.py` (requires `--truth-dir` and/or `--reco-file`).

| Flag                     | Default               | Meaning                                                                |
|--------------------------|-----------------------|------------------------------------------------------------------------|
| <code>--dict-path</code> | CoreUtils' built dict | Path to the built ROOT dictionary                                      |
| <code>--truth-dir</code> | none (skip)           | Dir of truth files; needs <code>--run</code> , <code>--n-events</code> |
| <code>--reco-file</code> | none (skip)           | Path to one Batch-TP0Recevent_*.root file                              |

## 6. A typical end-to-end run

```
source setup.sh
cmake -S . -B build -DCMAKE_BUILD_TYPE=RelWithDebInfo && cmake --build build -j

python3 run_faserps.py --n-events 500 # simulate
python3 run_batchreco.py --run 10000 --mask numuCC # reconstruct numuCC
build/bin/evDisplay.exe 10000 numuCC # look at the result
```

or, to exercise the whole pipeline and check it against the committed baseline:

```
python3 run_regression_tests.py
```

## 7. Tests

```
ctest --test-dir build --output-on-failure  
ctest --test-dir build -R Magnet --output-on-failure # run a subset by name
```

- FASERCAL's own gtest suite (MuonSpectrometerFieldTest, MagnetGeometryProbeTest, ...) builds automatically whenever FASER\_BUILD\_TESTS and FASER\_BUILD\_GEANT4\_PACKAGES are both ON (the default)
- the heavier simulate-reconstruct-compare regression suite is run\_regression\_tests.py, described above and in docs/REGRESSION\_TESTS.md

# Adding a new test

- 1 write `Tests/MyNewTest.cc`:

```
#include <gtest/gtest.h>
TEST(MySuite, MyCase) { EXPECT_EQ(2 + 2, 4); }
```

- 2 add a matching `add_executable()` / `target_link_libraries()` / `add_test()` block to `Tests/CMakeLists.txt`, modeled on `MuonSpectrometerFieldTest` / `MagnetGeometryProbeTest` — there's no automatic `*Test.cc` discovery, each test needs its own block
- 3 if it links `Geant4`, it also needs the `find_package(Geant4 ...)` already at the top of that file, plus the same `-Wl,--allow-shlib-undefined` Linux workaround the existing `Geant4`-linking tests use
- 4 reconfigure once (`cmake -S . -B build && cmake -build build -j, or fb`) so CMake picks up the new target — `ctest` then finds it automatically

Full walkthrough: `docs/HOWTO.md`, §7.

- docs/INSTALL.md — the full CMake build: prerequisites, every option, troubleshooting a clean build
- docs/GEANT4\_INSTALL.md — building/installing Geant4 itself
- run\_faserps.md / run\_batchreco.md — prose documentation for those two wrappers
- docs/README\_MuonDIS.md — MuonDIS physics and every /physics/muondis/... option
- docs/REGRESSION\_TESTS.md — the regression suite's full design, what's verified vs. not, and open items